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Resource

The Somatic Genomic Landscape of Glioblastoma

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SUMMARY

We describe the landscape of somatic genomic alterations based on multidimensional and comprehensive characterization of more than 500 glioblastoma tumors (GBMs). We identify several novel mutated genes as well as complex rearrangements of signature receptors, including *EGFR* and *PDGFRA*. *TERT* promoter mutations are shown to correlate with elevated mRNA expression, supporting a role in telomerase reactivation. Correlative analyses confirm that the survival advantage of the proneural subtype is conferred by the G-CIMP phenotype, and *MGMT* DNA methylation may be a predictive biomarker for treatment response only in classical subtype GBM. Integrative analysis of genomic and proteomic profiles challenges the notion of therapeutic inhibition of a pathway as an alternative to inhibition of the target itself. These data will facilitate the discovery of therapeutic and diagnostic target candidates, the validation of research and clinical observations and the generation of unanticipated hypotheses that can advance our molecular understanding of this lethal cancer.

INTRODUCTION

Glioblastoma (GBM) was the first cancer type to be systematically studied by The Cancer Genome Atlas Research Network (TCGA). The initial publication (TCGA, 2008) presented the results of genomic and transcriptomic analysis of 206 GBMs, including mutation sequencing of 600 genes in 91 of the samples. The observations provided a proof-of-concept demonstration that systematic genomic analyses in a statistically powered cohort can define core biological pathways, substantiate anecdotal observations, and generate unanticipated insights.

The initial publication reported biologically relevant alterations in three core pathways, namely p53, Rb, and receptor tyrosine kinase (RTK)/Ras/phosphoinositide 3-kinase (PI3K) signaling (TCGA, 2008). Efforts to link the alterations found in these pathways to the distinct molecular and epigenetic subtypes of glioblastoma revealed that coordinated combinations were enriched in different molecular subtypes, which may affect clinical outcome and the sensitivity of individual tumors to therapy (Noushmehr et al., 2010; Verhaak et al., 2010).

Above and beyond these observations, it has become evident that GBM growth is driven by a signaling network with functional redundancy that permits adaptation in response to targeted molecular treatments. Thus, a comprehensive catalog of molecular alterations in GBM, based on multidimensional high-resolution data sets, will be a critical resource for future investigative efforts to understand its pathogenesis mechanisms, inform tumor biology and ultimately develop effective therapies against this deadly cancer.

Toward those ends, TCGA has expanded the scope and depth of molecular data on GBM, including adoption of next-generation sequencing technology (TCGA, 2011, 2012a, 2012b, 2012c). Here, we report the efforts of the TCGA GBM Analysis Working Group (AWG) to further our understanding of GBM pathobiology by constructing a detailed somatic landscape of GBM through a series of comprehensive genomic, epigenomic, transcriptomic, and proteomic analysis.

RESULTS

Samples and Clinical Data

As summarized in Table 1, the data set contains molecular and clinical data for a total of 543 patients. Note that different subsets of patients were assayed on each technology platform. The most significant additions to the GBM data set include sequencing of GBM whole-genomes, coding exomes, transcriptomes, as well as profiling of expanded DNA methylomes and a targeted proteome. In particular, 291 pairs of germline-tumor native DNAs (e.g., without whole-genome amplification) were characterized by hybrid-capture whole-exome sequencing (WES) and of these, 42 pairs underwent deep coverage whole-genome sequencing (WGS). The transcriptomes of 164 RNA samples were profiled by RNA-sequencing (RNA-seq). Protein expression profiles were generated from 214 patient samples using reverse phase protein arrays (RPPA). The data package associated with this report was frozen on July 15, 2013 and is available at the Data Portal (https://tcga-data.nci.nih.gov/docs/publications/gbm_2013/).

TCGA sample collection spanned 17 contributing sites (Table S1). Tier 1 clinical data elements (including age, pathology, and survival) are available on 539 of 543 patients (99.6%) and tier 2 data including treatment information on 525 patients (96.7%) (Figure S1, available online, see Data Portal). Clinical characteristics of this patient cohort are similar to our previous report in 2008 (TCGA, 2008) with a median age of 59.6 years and a male to female ratio of 1.6 (333:209). Median overall survival was 13.9 months with 2 year survival of 22.5% and 5 year survival of 5.3%. Due to TCGA selection of primary GBM, *IDH1* mutation is infrequent in the TCGA cohort compared to other published series. Of the 423 patients with adequate sequencing coverage (by either whole-exome next-generation sequencing or previously reported Sanger-based sequencing), 28 (6%) had the *IDH1*-R132H mutation, whereas one individual had an R132G and one had an R132C mutation. No *IDH2* mutations were found. The associated G-CIMP methylation pattern was present in all cases of *IDH1* mutation (R132H/G/C), whereas seven G-CIMP cases lacked *IDH1* mutations. Overall, G-CIMP pattern was present in 42 out of 532 cases (7.9%). Clinically relevant *MGMT* DNA methylation status was estimated from CpG islands as previously described (Bady et al., 2012). Conventional positive prognostic factors were confirmed by univariate analysis: age < 50 (OS 21.9 versus 12.3 months, $p = 2.4 \times 10^{-11}$), *MGMT* DNA methylation (16.9 versus 12.7, $p = 0.0018$), *IDH1* mutation (35.4 versus 13.3, $p = 1.55 \times 10^{-5}$) and G-CIMP DNA methylation (38.3 versus 12.7, $p = 8.3 \times 10^{-9}$). Age, *MGMT*, and *IDH1*/G-CIMP status were independently significant in multivariate analysis (Table S1).

Patients in this TCGA cohort were diagnosed between 1989 and 2011, with 414 patients (76%) receiving their diagnosis in

Table 1. Characterization Platforms and Data Availability

Data Type	Platforms	Cases in 2008	Cases in 2013
DNA sequence of exome	Illumina on native DNA	0	291
	Sanger on native DNA	91	148
	Illumina on whole genome amplified DNA	0	163
DNA sequence of whole genome	Illumina on native DNA	0	42
DNA copy number/ genotype	Affymetrix SNP6	206	578
	Agilent 224K/415K	206	413
mRNA expression profiling	Affymetrix U133A	206	544
	Affymetrix Exon	201	417
mRNA sequencing	Illlumina on native cDNA	0	164
CpG DNA methylation	Illumina GoldenGate	242	242
	Illumina 27K	0	285
	Illumina 450K	0	113
miRNA expression profiling	Agilent	205	491
Protein expression profiling	Reverse phase protein arrays	0	214
Clinical characteristics	Tier 1/Tier 2	206	543

or after 2002 when the use of concurrent temozolomide (TMZ) with adjuvant radiation became widely adopted. Combined TMZ chemotherapy and radiation treatment is documented for 40% of all patients (217/543), and for 50.2% of the 414 patients diagnosed in or after 2002. Summaries of treatment classification classes are provided in Supplemental Information.

Whole-Exome Sequencing Identifies Significantly Mutated Genes in Glioblastomas

Solution-phase hybrid capture and whole-exome sequencing were performed on paired tumor and normal native genomic DNA obtained from 291 patients. Overall, 138-fold mean target coverage was achieved, with 92% of bases covered at least 14-fold in the tumor and 8-fold in the normal—the threshold which offers 80% power to detect mutations with an allelic fraction of 0.3 (Carter et al., 2012) (see Extended Experimental Procedures). Overall, of the 291 tumor exomes sequenced, 21,540 somatic mutations were identified, with a median rate of 2.2 coding mutations per megabase (lower-upper quartile range, 1.8–2.3). Among the somatic mutations were 20,448 single-nucleotide variants (SNVs), 39 dinucleotide mutations, and 1,153 small insertions and deletions (indels). The SNVs mutations included 5,379 silent, 3,901 missense, 831 nonsense, 360 splice-site and 760 mutations resulting in a frame shift.

Mutations were evaluated across samples to distinguish genes which appear targeted by driver rather than passenger mutations using both MutSig (TCGA, 2008, 2011, 2012a, 2012b, 2012c) and InVEx algorithms (Hodis et al., 2012). MutSig assesses mutation significance as a function of gene size, trinucleotide context, gene structure, and background mutation rates. InVEx compares the ratio of nonsilent exonic mutations to synonymous and intronic/UTR nucleotide variants, an algorithm that is particularly effective for genomes with elevated mutation rates such as melanoma and lung adenocarcinoma. When both InVEx and MutSig algorithms were run on the same data set, a total of 71 genes were identified as significantly mutated genes (SMG). To validate mutation calls, all 757 SNVs and indels detected by exome sequencing in these 71 SMGs were subject to orthogonal validation by targeted resequencing in 259 tumor/normal pairs. At sites with adequate coverage to detect the mutant alleles, 98% of SNVs, 84% of insertions, and 82% of deletions were validated (see Extended Experimental Procedures).

As summarized in Figure 1A, both InVEx and MutSig algorithms identified previously reported genes as significantly mutated in GBM, namely *PTEN*, *TP53*, *EGFR*, *PIK3CA*, *PIK3R1*, *NF1*, *RB1*, *IDH1*, and *PDGFRA* (Figure 1A). In addition, both algorithms identified the leucine-zipper-like transcriptional regulator 1 (*LZTR1*), mutated in ten samples, as a novel significantly mutated gene in GBM (Table S2, Figure S2). *LZTR1*, a putative transcriptional regulator associated with the DiGeorge congenital developmental syndrome (Kurahashi et al., 1995), has not previously been implicated in cancer. It is located at chromosome 22q, and in five of six samples with available copy-number data it was simultaneously targeted by hemizygous deletion.

MutSig additionally identified 61 additional genes (71 overall) with mutation frequency above background with a q-value of < 0.1 (Table S2). These included spectrin alpha 1 (*SPTA1*, mutated in 9%), which encodes a cell motility protein that interacts with the ABL oncogene and is related to various hereditary red blood cell disorders; *ATRX* (6%), a member of the SWI/SNF family of chromatin remodelers recently implicated in pediatric and adult high-grade gliomas (Kannan et al., 2012; Liu et al., 2012; Schwartzentruber et al., 2012); *GABRA6* (4%), an inhibitory neurotransmitter in the mammalian brain; and *KEL* (5%), which codes for a transmembrane polymorphic antigen glycoprotein (Figure S2). Albeit at low frequency, several hotspot mutations were found to be significant in this cohort of GBM, most notably the *IDH1* R132H mutation. The *BRAF* V600E sequence variant, which confers sensitivity to vemurafenib in melanoma (Chapman et al., 2011a), was detected in five of 291 GBMs (1.7%). Mutation of H3.3 histones, reported in pediatric gliomas (Schwartzentruber et al., 2012), were not observed in this cohort of primary GBM.

To facilitate exploration of mutation data by noncomputational biologists, we developed a patient-centric table (PCT) that categorizes each gene in each sample by the type of mutation (silent, missense, InDel, etc.) observed, and describes the confidence of each call based on the coverage in normal and tumor samples (see Data Portal, Extended Experimental Procedures). To illustrate one potential use of this table, we interrogated the mutation pattern of 161 genes functionally linked to chromatin organization (hereafter referred to as CMG or "chromatin modification genes," see Extended Experimental Procedures) using this PCT. In total, 135 samples or 46% of the sample cohort harbored at least one nonsynonymous mutation in this CMG gene set (Figure 1B). Importantly, CMG mutations were found to be mutually

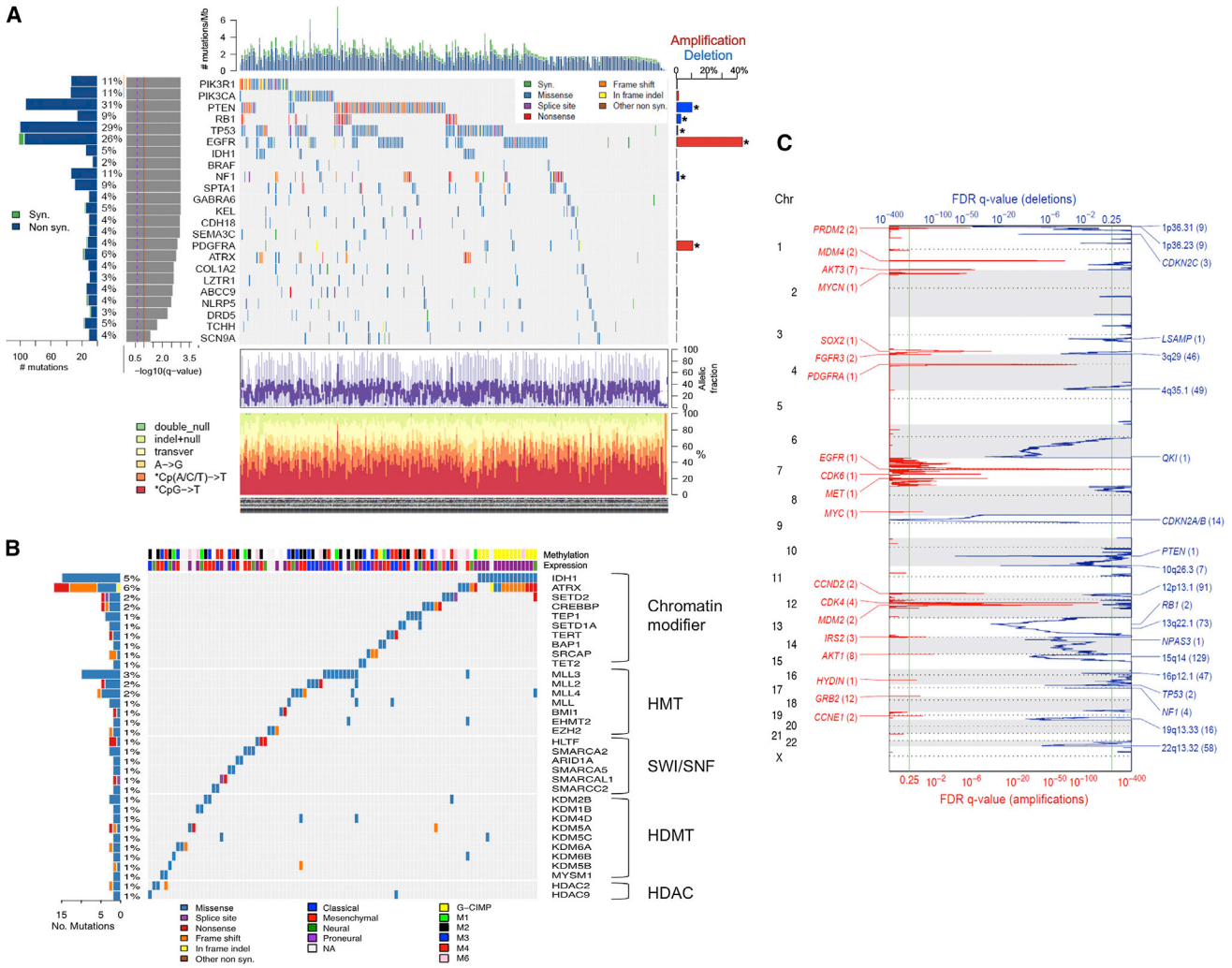


Figure 1. Somatic Genomic Alterations in Glioblastoma

(A) Summary of significantly mutated genes from 291 exomes. Specific mutations for LZTR1, SPTA1, KEL, and TCHH are shown in Figure S2. a: Number of mutations per sample (substitutions and indels). b, rate of mutations per gene and percentage of samples affected. Central heat map: Distribution of significant mutations across sequenced samples, color coded by mutation type. c: Overall count and significance level of mutations as determined by log(10) transformation of the MutSig q-value. Red line indicates a q-value of 0.05. d: Summary of focal amplifications (red) and deletion (blue) determined from DNA copy-number platforms (asterisk denotes inclusion in statistically significant recurrent CNA by GISTIC). e: Average fraction of tumor reads versus total number of reads per sample. f: top, rates of nonsilent mutations within categories indicated by legend; bottom, mutation spectrum of somatic substitutions of samples in each column.

(B) Mutations in 38 genes related to specific epigenetic function categories (out of 161 genes linked to chromatin modification) across 98 GBMs (out of 292 GBM). *IDH1* mutation status is included to illustrate its co-occurrence with *ATRX* mutation. An additional 37 GBMs harbored mutations in one of the remaining 129 CMGs.

(C) Recurrent sites of DNA copy-number aberration determined from 543 samples by the GISTIC algorithm. Statistically significant, focally amplified (red) and deleted (blue) regions are plotted along the genome. Significant regions (FDR < 0.25) are annotated with the number of genes spanned by the peak in parentheses. For peaks that contain a putative oncogene or tumor suppressor, the gene is noted.

exclusive overall by MEMo analysis (p = 0.0008) (Ciriello et al., 2012), suggesting potential biological relevance of chromatin modification in GBM.

Genomic Gains and Losses in GBM

We expanded our previously reported DNA copy-number analysis from 206 GBMs (TCGA, 2008) to 543 samples. The larger data set, coupled with improvement of the analytical algorithm GISTIC (Mermel et al., 2011), resulted in a significant refinement of previously defined amplification and deletion peaks, thus allowing improved nomination of candidate gene targets for several recurrent somatic copy-number aberrations (SCNA) (Figure 1C). The most common amplification events on chromosome 7 (*EGFR*/*MET*/*CDK6*), chromosome 12 (*CDK4* and *MDM2*), and chromosome 4 (*PDGFRA*) were found at higher frequencies than previously reported (Table S3), and often contained only a single gene in the common overlapping region. Additionally, frequent gains of genes such as *SOX2*, *MYCN*, *CCND1*, and

CCNE2 were precisely established. Except for the highly recurrent homozygous deletions in *CDKN2A/B,* all statistically significant DNA losses were hemizygous. Losses were more frequent than amplifications, as has been reported as a general pattern in cancer (Beroukhim et al., 2010). We were able to pinpoint single genes as deletion targets in some cases, most notably in recurrent deletion of 6q26. Although the 6q26 deletion has previously been associated with other candidates such as *PARK2,* our analysis unequivocally defined *QKI* as the sole gene within the minimal common region and the target of homozygous deletion in nine cases. The *QKI* gene was also mutated in five cases without evidence of deletion (two frame-shift, two missense, and one splice-site mutation). This is consistent with a recent publication demonstrating that *QKI* functions as a tumor suppressor in GBM by acting as a p53-responsive regulator of mature miR-20a stability to regulate TGFβR2 expression and TGFβ network signaling (Chen et al., 2012). Other single gene deletion targets include *LRP1B*, *NPAS3, LSAMP*, and *SMYD3*. Similar to the mutation data, we have also algorithmically generated a patient-centric table summarizing DNA copy-number aberration and DNA methylation status for each gene and miRNA for each of the cases in the cohort (see Data Portal).

Recurrent Structural Rearrangements Defined by Genomic and Transcriptomic Sequencing

To explore genomic and transcriptomal structural rearrangements, we performed whole-genome paired-end sequencing with deep coverage on 42 pairs of tumor and matched germline DNA samples as well as RNA sequencing (RNA-seq) of 164 GBM transcriptomes (Table S4). We detected genomic rearrangements using BreakDancer and BamBam (Sanborn et al., 2013) (see Extended Experimental Procedures), in addition to expressed RNA fusions using PRADA (http://sourceforge.net/projects/prada/). In total, we identified 238 high-confidence candidate somatic rearrangements, including 49 interchromosomal, 125 intrachromosomal, and 64 intragenic structural variants (Figures 2A and 2B; Table S4). The number of events per sample ranged from 0 to 32 (median: 2), with one sample containing a distinctively high number of rearrangements in the context of local chromothripsis involving a 7.5 Mb region on chromosome 1. No rearrangements were detected in eight samples. Overall, the number of rearrangements generally appeared lower than what has been previously reported for prostate cancer (Sanborn et al., 2013), lung adenocarcinoma (Imielinski et al., 2012), and melanoma (Berger et al., 2012). Recurrent intragenic events were detected in seven genes: *EGFR* (n = 12), *CPM* (n = 3), *PRIM2* (n = 3), *FAM65B* (n = 2), *PPM1H* (n = 2), *RBM25* (n = 2), and *HOMER2* (n = 2). Because unbalanced structural rearrangements in DNA can be detected as breakpoints in DNA copy-number profiles, we investigated whether CNA breakpoints could indicate potential sites of recurrent structural rearrangement using all 492 samples with aCGH data (n = 492). Of note, 41 of 129 high-confidence rearrangement events from whole-genome sequencing (WGS) involved genes identified as significant targets of recurrent intragenic copy-number breakpoints (iCNA) in the larger cohort of GBM based on DNA copy-number profiles (Table S4, Data Portal).

RNA seq analysis identified 48 interchromosomal and 180 intrachromosomal mRNA fusion transcripts in 106 of 164 samples (Figure 2C; Table S4). Approximately 37% of these were in-frame transcripts, 35% were out-of-frame, and the remaining 29% were involved a 3′ or 5′ untranslated region (Figure S3A). A substantial portion (44%) of the intrachromosomal events resulted from recombination of genomic loci located less than 1 Mb apart. A notable example is the recently reported oncogenic *FGFR3-TACC3* inversion (Singh et al., 2012), which was detected in two cases. Interestingly, the *FGFR3/TACC3* locus was focally amplified in both samples, suggesting that CNA could serve as a marker of *FGFR3-TACC3* rearrangement. Overall, focal amplifications involving *FGFR3* or *TACC3* were detected in 14 of 537 GBM copy-number profiles (2.6%).

Ten of the 42 GBMs with WGS analyses demonstrated rearrangements between *EGFR* and adjacent genes such as *BRIP* (n = 2) and *VOPP1* (n = 2), or structural variants of genes surrounding the *EGFR* locus, such as *LANCL2* and *PLEXHA* (n = 2) (Table S4). Both types of 7p11 rearrangements were detected in six samples. This pattern was confirmed in the RNA-seq data where eighteen samples of 164 samples showed evidence of transcribed fusion transcripts, such as *EGFR-SEPT14* (n = 6), *SEC61G-EGFR* (n = 4), *LANCL2-SEPT14* (n = 1), and *COBL-SEPT14* (n = 1) (Table S4). These fusions tended to be part of a focal gain, suggesting a complex rearrangement (Figure S3B).

Genomic rearrangements pertaining to chromosome arm 12q were identified in 11 of 42 whole genomes and 12q-associated fusion transcripts were found expressed in 25 of 164 transcriptomes. A variety of different genomic and transcriptomic variants were found on 12q though none were recurrent (Table S4). The majority of 12q lesions occurred in tandem, i.e., as adjacent events in the same GBM. As an illustration, a single sample showed a pattern in which 15 nonadjacent segments (14 from chromosome 12 and one fragment from chromosome 7) were highly amplified (>40 copies) with eight 12q rearrangement events, including the *MDM2*, *CDK4*, and *EGFR* oncogenes (Figure S3C). WGS analysis reconstructed two independent circular paths that accounted for all of the amplified segments (Figure S3C). Each circle contained at least one oncogene, with one circle (0152-DM-A) containing one copy of *CDK4* and two copies of *MDM2* and the other circle (0152-DM-B) containing one copy of *EGFR*. These reconstructed circles are most consistent with extrachromosomal double minute chromosomes (Kuttler and Mai, 2007). Recently, the same data set was used to identify enrichment of genomic breakpoints relating to chromosome 12q14-15, a locus harboring the *MDM2* and *CDK4* oncogenes, associated with less favorable outcome (Zheng et al., 2013), and the reconstruction of double minutes confirmed using orthogonal methods (Sanborn et al., 2013).

EGFR Is Frequently Targeted by Multiple Alterations of DNA and RNA

As anticipated, *EGFR* was among the most frequently mutated genes and RNA-seq detected a diversity of altered transcripts (Figure 3A). *EGFR* mutations were accompanied by regional DNA amplification in the majority of cases, leading to a wide range of mutation allelic frequencies. Comparing the allelic frequencies of point mutations in DNA- and RNA-seq data revealed

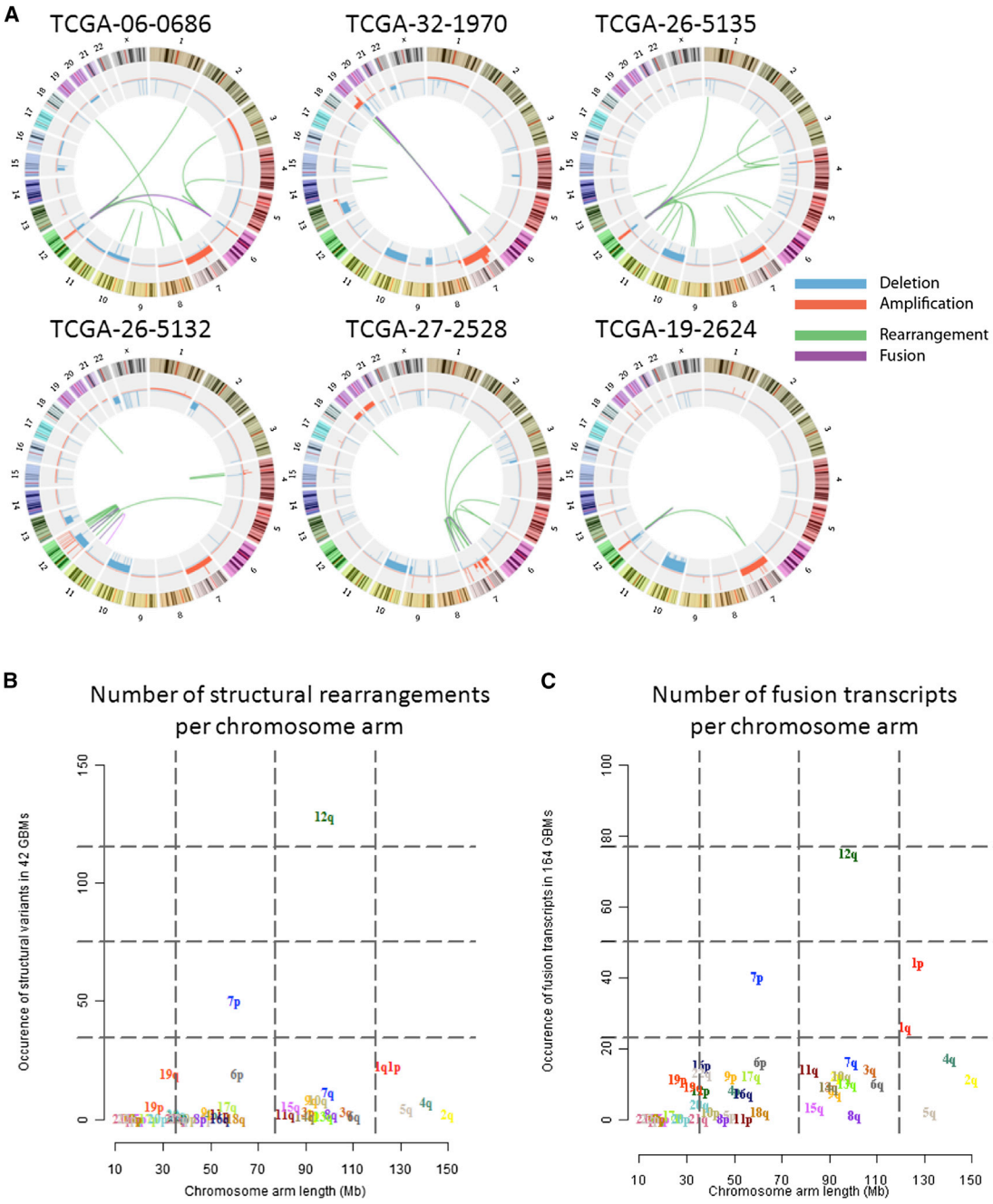


Figure 2. Structural Rearrangements and Transcript Variants in GBM

(A) Circos plots of structural DNA and RNA rearrangements in six GBMs, selected from 28 cases with available whole-genome and RNA sequencing based on their rearrangement frequency. Outer ring indicates chromosomes. Copy-number levels are displayed along the chromosome map in red (copy-number gain) and blue (copy-number loss). Each line in the center maps a single structural variant to the site of origin for both genes (see Figure S3 for additional analysis of fusion transcripts derived from RNA sequencing).

(B) The chromosome arm of origin of both ends of each rearrangement detected in whole-genome sequencing data from 42 GBM were counted and compared to chromosome arm length.

(C) The chromosome arm of both partners in fusion transcripts detected from RNA sequencing data from 164 GBM were counted and compared to chromosome arm length.

a high degree of concordance between the type and prevalence of mutations at the DNA level and the composition of expressed mRNA transcripts (Figure S4A).

RNA-seq also provided a complete picture of aberrant exon junctions and a semiquantitative assessment of their expression levels. Transcript allelic fraction (TAF) was calculated

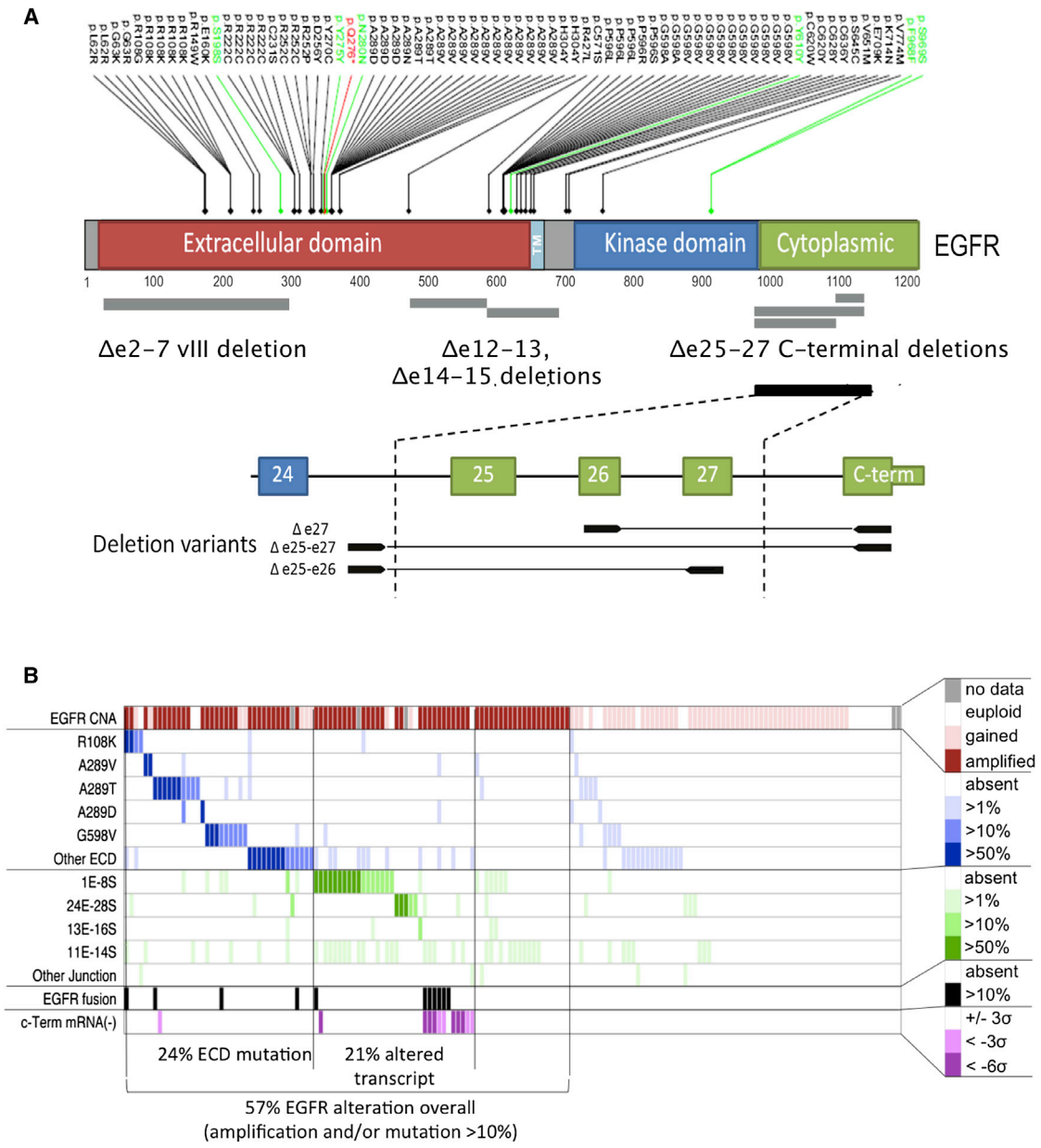


Figure 3. Somatic Alterations of the EGFR Locus

(A) EGFR protein domain structure with somatic mutations summarized from 291 GBMs with exome sequencing and transcript alterations identified across 164 GBMs with RNA sequencing.

(B) EGFR alterations are summarized by transcript prevalence in 164 GBMs with RNA sequencing. Red, top: focal amplification or regional gain inferred from DNA copy number. Blue, Prevalence of sequencing reads with EGFR point mutation. Green, prevalence of reads with aberrant exon-exon junctions (e.g., 1E-8S is a junction spanning from the end of exon 1 to the start of exon 8, consistent with EGFRvIII mutation). Black, EGFR fusion transcript detected (see rearrangements). Purple: C-terminal deletion inferred from relative under expression of C-terminus exons 27-29 vs. kinase domain exons by >3 or >6 SD. See related Figure S4 for comparison of EGFR mutations in DNA and RNA and for a summary of EGFR rearrangements.

as the ratio of each aberrant exon junction to the sum of aberrant and wild-type junctions at the 3′ junction end, corrected for read depth (80% confidence, binomial confidence interval). TAFs for recurrent point mutations and junctions are summarized in Table S5. In 11% of tumors, the aberrant exon 1–8 junction characteristic of *EGFRvIII* was highly expressed (≥10% TAF), whereas 19% showed at least a low level expression (≥1%). The results were concordant with an independent assessment of *EGFRvIII* by digital mRNA assay using barcoded probes (nCounter, Nanostring Technologies and by real-time PCR; see Data Portal). Although the biological or clinical relevance of low-level *EGFRvIII* expression remains

to be demonstrated, *EGFRvIII* expression in a minor population of GBM cells has been shown to confer a more aggressive tumor phenotype through paracrine mechanisms (Inda et al., 2010).

A variety of other recurrent noncanonical *EGFR* transcript forms were detected in the RNA-seq data (Figures 3A and S4B). Three different C-terminal rearrangements targeting the cytoplasmic domain of the *EGFR* were detected at ≥ 10% TAF in 3.7% of cases and at ≥ 1% TAF in another 9%. Comparison with WGS data confirmed the presence of C-terminal deletions in nine cases where sequence data were available. C-terminal deletion variants have previously been associated with gliomagenesis in experimental rodent systems in vivo (Cho et al., 2011). The prevalence of *EGFR* C-terminal deletion reported here is likely an underestimate since complete loss of the C terminus may yield aberrant terminal junctions not mappable by transcriptome sequencing. Relative underexpression of C terminus exons 27–29 (<3 SD) was readily apparent in another 7.3% of cases without detectable aberrant junctions (Figure 3B).

We identified two relatively uncharacterized recurrent *EGFR* variants, namely deletions of exons 12–13 (Δ12–13) in 28.7% and exons 14–15 (Δ14–15) in 3%. *EGFR* Δ12–13 has been previously identified by RT-PCR analysis of glioma (Callaghan et al., 1993). Both Δ12–13 and Δ14–15 appear to be expressed in minor allelic fractions (<10%), raising the question of whether they result from splicing aberration or genomic deletion. Among tumors expressing Δ12–13 mRNA, analysis of aberrant junctions in WGS data (BamBam) failed to identify concordant DNA deletion in 14/15 cases where data were available. One case showed a concordant breakpoint as a minor component of a highly rearranged locus. By comparison, *EGFRvIII*-expressing tumors had concordant deletion spanning exons 2–7 in all seven cases where WGS data were available (Table S5).

In total, 38.4% of cases harbored an *EGFR* genomic rearrangement or a point mutation expressed in at least 10% of transcripts (Figure 3B; Table S5). Overall, 57% of GBM showed evidence of mutation, rearrangement, altered splicing, and/or focal amplification of *EGFR*. Whereas *PDGFRA* showed no recurrent gene fusions, intragenic deletion of exons 8 and 9 (*PDGFRA* Δ8,9) was highly expressed (≥10% TAF) in 1 of the 164 samples with RNA sequencing data. Low-level expression of *PDGFRA* Δ8,9 was far more prevalent in the RNA-seq data (n = 29 of 163) and could represent a splice variant. This result is concordant with previously reported estimates of Δ8,9 expression (Ozawa et al., 2010). A novel *PDGFRA* variant with deletion of exons 2–7 was found highly expressed in a single case (TCGA-28-5216).

The Landscape of Somatic Alterations in Glioblastoma

The addition of whole-exome and transcriptomal sequencing data have extended the palette of somatic alterations affecting major cancer pathways in GBM. Figure 4 presents a landscape view of the canonical signal transduction and tumor suppressor pathways in GBM based on whole-exome sequencing data of 291 patients. Unsupervised analysis of 251 GBMs with both copy number and WES mutation data identified genes sets (modules) in which somatic alterations were significantly mutually exclusive (MEMo; Ciriello et al., 2012). This analysis confirmed mutual exclusivity among alterations affecting the p53 pathway (*MDM2*, *MDM4*, and *TP53*), the Rb pathway (*CDK4*, *CDK6*, *CCND2*, *CDKN2A/B*, and *RB1*), and various components influencing the PI3K pathway (*PIK3CA*, *PIK3R1*, *PTEN*, *EGFR*, *PDGFRA*, and *NF1*) (Table S6).

As shown, at least one RTK was found altered in 67.3% of GBM overall: *EGFR* (57.4%), *PDGFRA* (13.1%), *MET* (1.6%), and *FGFR2/3* (3.2%). Nearly half of the tumors with focal amplification and/or mutation of *PDGFRA* harbored concurrent *EGFR* alterations (42.4%, 14/33), as did the majority of *MET*-altered tumors (3/4), reflecting a pattern of intratumoral heterogeneity that has been previously documented by in situ hybridization (Snuderl et al., 2011; Szerlip et al., 2012).

PI3-kinase mutations were found in 25.1% of GBM (63/251), with 18.3% affecting p110alpha and/or p85alpha subunits and 6.8% in other PI3K family genes. *PI3K* mutations were mutually exclusive of *PTEN* mutations/deletions ($p = 0.0047$, Fisher's exact), with 59.4% of GBM showing one or the other (149/251). Considering the RTK genes, PI3-kinase genes, and *PTEN*, 89.6% of GBM had at least one alteration in the PI3K pathway and 39% had two or more. The *NF1* gene was deleted or mutated in 10% of cases, and never co-occurred with *BRAF* mutations (2%).

Concordant with the previous TCGA GBM report, the p53 pathway was found to be dysregulated in 85.3% of tumors (214/251), through mutation/deletion of *TP53* (27.9%), amplification of *MDM1/2/4* (15.1%), and/or deletion of *CDKN2A* (57.8%). As expected, *TP53* alterations were mutually exclusive with amplification of *MDM* family genes ($p = 0.0003$) and *CDKN2A* ($p = 1.99 \times 10^{-7}$). Concurrently, 78.9% of tumors had one or more alteration affecting Rb function: 7.6% by direct *RB1* mutation/deletion, 15.5% by amplification of *CDK4/6*, and the remainder via *CDKN2A* deletion.

As reported for lower grade gliomas (Ichimura et al., 2009), 12 of the 13 GBMs with *IDH1* hotspot mutations harbored concurrent *TP53* mutations. Consistent with recent reports, mutations in SWI/SNF complex gene *ATRX* often co-occurred in these cases (Figure 4B). Mutations in *IDH1* and *ATRX* appear to be more prevalent in GBM samples without RTK alteration ($p = 7.2 \times 10^{-5}$ and 7.3×10^{-4}, respectively), tumors genotypically more consistent with secondary GBM (Ohgaki and Kleihues, 2007).

Telomerase reverse transcriptase (*TERT*) promoter mutations were recently reported in glioma, mapping to positions 124 (C228T) and 146 bp (C250T) upstream of the *TERT* ATG start site (Killela et al., 2013). Of the 42 cases with deep coverage WGS data, 25 samples had adequate coverage (read count > 10) of the *TERT* promoter for mutational analysis. We detected the C228T mutation in 15 of the 25 cases, whereas the C250T variant was found in another six cases (Figure 4C). *TERT* promoter mutations at these two hot spots were correlated with upregulated *TERT* expression at the RNA level (Figure 4C). Interestingly, the four GBMs with nonmutated *TERT* promoters all harbored *ATRX* mutations and these were concurrent with *IDH1* and *TP53* mutations as recently described (Liu et al., 2012). Finally, in line with the role of ATRX in alternative lengthening of telomeres (ALT) (Lovejoy et al., 2012), *ATRX* mutant GBM tumors do not exhibit elevated *TERT* RNA expression compared to tumors with *TERT* promoter mutations (Figure 4C). Taken together, these data suggest that maintenance of the telomere either through reactivation of telomerase by *TERT*

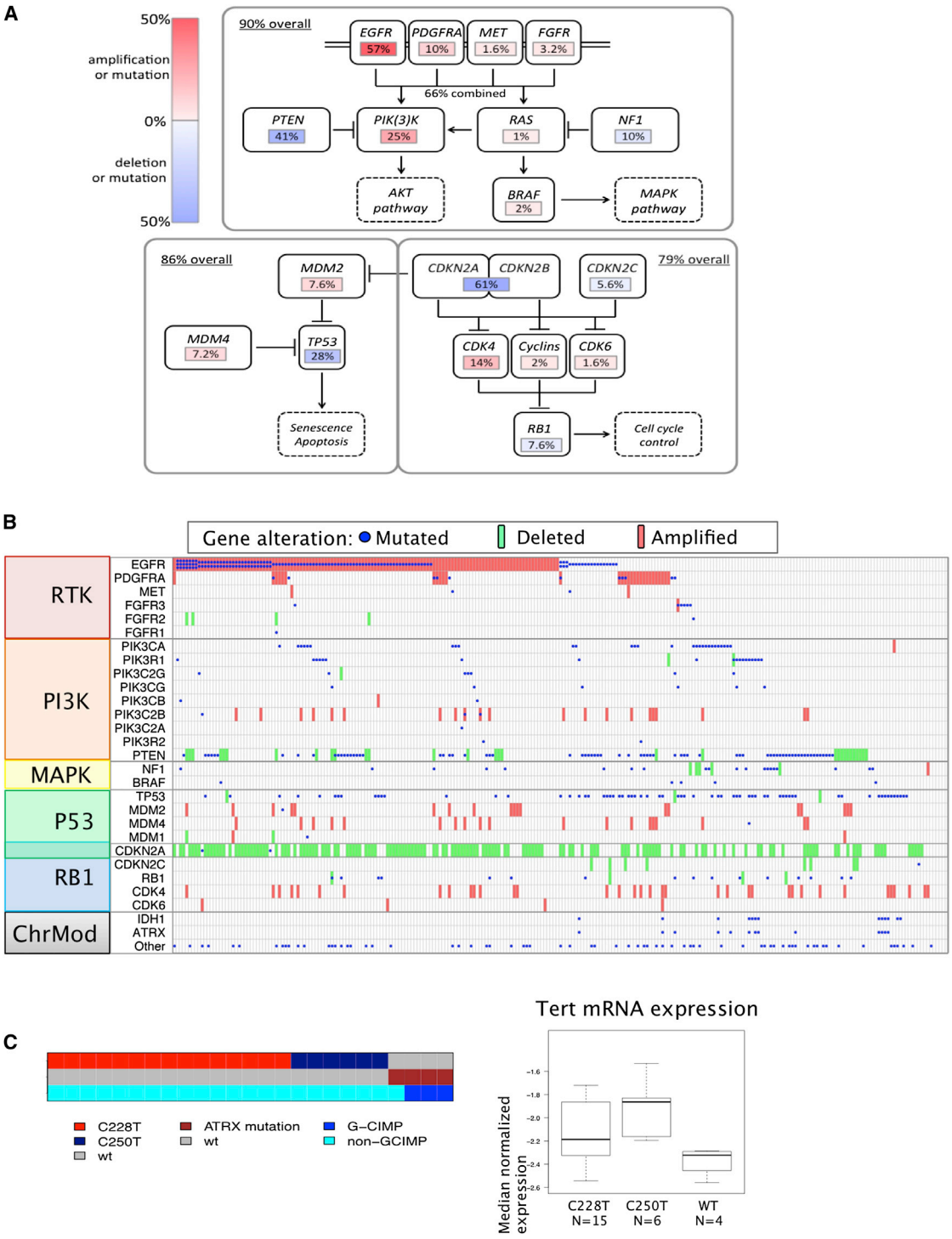


Figure 4. Landscape of Pathway Alterations in GBM

Alterations affecting canonical signal transduction and tumor suppressor pathways are summarized for 251 GBM with both exome sequencing and DNA copy-number data. Rearrangements are underestimated in this summary since RNA-seq data were available for only a subset of cases with exome sequencing data (153/291, 61%).

(A) Overall alteration rate is summarized for canonical PI3K/MAPK, p53 and Rb regulatory pathways.

(B) Per-sample expansion of alterations summarized in 5A. Mutations (blue), focal amplifications (red), and homozygous deletions are selected from the patient-centric tables and organized by function. All missense, nonsense and frame-shift mutations are included. EGFRvIII is inferred from RNA data and included as a

(legend continued on next page)

promoter mutation-induced increased *TERT* expression or ALT as a result of *ATRX* mutation is a requisite step in GBM pathogenesis.

Although reported median survival for patients with GBM ranges from 12–18 months, a subset of individuals will survive for more than 3 years (Dolecek et al., 2012; Dunn et al., 2012). We cross-referenced our data set to identify any factor(s) associated with long-term survival (n = 39 or 7.7% of the cohort). Although no specific genomic alteration was significantly overrepresented in this subset, amplifications of *CDK4* and *EGFR* and deletion of *CDKN2A* were observed at decreased frequencies in these long survivors (see Data Portal). Age at diagnosis was found to be a major determinant, with 79% of long-term survivors being diagnosed at younger than 50 years of age. Despite their relatively favorable prognosis, only one third of patients with G-CIMP+ GBM survived beyond 3 years, suggesting that other factors yet to be identified are contributing to overall long-term survival of GBM patients.

Molecular Subclasses Defined by Global mRNA Expression and DNA Methylation

Widespread differences in gene expression have previously been reported in GBM, grouping TCGA tumors into proneural, neural, classical, and mesenchymal transcriptomic subtypes (Phillips et al., 2006; Verhaak et al., 2010). Samples not included in previously published analysis (n = 342) were classified into one of classes using single-sample gene set enrichment analysis (Figure 5A and Table S7) Similarly, we sought to assign each case in the TCGA cohort to one of the DNA methylation subclasses. The promoter DNA methylation array platforms used by TCGA have evolved with increasing resolution from the Illumina GoldenGate (n = 238), Infinium HumanMethylation27 (HM27, n = 283) and Infinium HumanMethylation450 (HM450, n = 76) platforms (Figure S5A). We reanalyzed a total of 396 GBM samples, comprised of 305 new GBM samples profiled on the HM27 (n = 192) and HM450 (n = 113) platforms in addition to 91 cases profiled on HM27 that were included previously (Noushmehr et al., 2010). Hierarchical consensus clustering of the DNA methylation profiles stratified these 396 GBM cases into six classes, including G-CIMP (Figures 5B, S5B and S5C, and Table S7). Cluster M1 (35/58, 60%) is enriched for mesenchymal GBMs while cluster M3 (18/31, 58%) is enriched for classical subtype (Figure 5B, red and blue, respectively). As expected, the G-CIMP cluster is enriched for proneural subtype tumors.

To be able to perform more robust exploration of the relationship of G-CIMP phenotype to other genomic alterations, we incorporated the previously reported G-CIMP status (Noushmehr et al., 2010) on 175 additional GBM cases profiled on the GoldenGate platform. A total of 534 GBM cases were used in the following integrative analyses. The age of GBM diagnosis was statistically different (41 year versus 56 year; p value = 0.008) between proneural G-CIMP (n = 28) and proneural non-G-CIMP (n = 22) subtypes, reinforcing the notion that the epigenomics of these transcriptomically similar patients mark distinct etiologies and/or disease characteristics. We observed seven G-CIMP(+) cases lacking *IDH1* mutation. These were similar to G-CIMP cases harboring *IDH1* mutations with respect to their median age at diagnosis (40 year versus 37 year, p value = 0.58) and overall survival (mean 913 days versus 1,248 days, p value = 0.45). *IDH2* mutation was not detected in these seven G-CIMP+/*IDH1* wild-type GBM, suggesting that alternative pathway(s) responsible for the hypermethylator phenotype.

Next, to identify genomic alterations enriched in each of the transcriptomic or epigenomic subtypes, we referenced the patient-centric tables to count DNA mutation and copy-number aberration events per subtype. This analysis confirmed previous reports, demonstrating significant associations between *PDGFRA* amplification and the non-G-CIMP+ proneural subgroup, as well as *NF1* inactivation and the mesenchymal subtype (Figure 5A). Additionally, the enhanced power of the larger data set identified an enrichment of *ATRX* mutations and *MYC* amplifications in the G-CIMP+ subtype, *CDK4* and *SOX2* amplifications in proneural subtype, and broad amplifications of chromosomes 19 and 20 in the classical subtype (Figure 5A). In contrast to G-CIMP, cluster M6 was relatively hypomethylated, with a predominance of nonmutated *IDH1* cases belonging to the proneural subtype (22/37, 59%) with concurrent *PDGFRA* amplification (Figure 5B).

To explore a plausible connection between chromatin deregulation and DNA methylation, we counted mutations in the 161 CMGs (Figure 1B) per each methylation subclass. In addition to the association of *IDH1* and *ATRX* mutations and G-CIMP, mutations of other CMGs were enriched across the M2, M4, and M6 subclasses (38% of cases in these three subclasses harbor at least one CMG mutation versus 18% among the other classes, p = 0.0015). Furthermore, cases with missense mutation or deletion of *MLL* genes (n = 18) or *HDAC* family genes (n = 4) clustered in the M2 DNA methylation subtype (10/21). These patterns of co-occurrence suggest a functional relationship between chromatin modification and DNA methylation that remains to be elucidated. Recently, Sturm et al. reported that adult and pediatric GBM with alterations of *IDH1*, *H3F3A*, and receptor tyrosine kinases (RTK) were associated with epigenetic subtypes (Sturm et al., 2012). We compared the Sturm et al. methylation-based classification with ours using the 74 TCGA cases that were also classified by those authors. We found that four tumors classified as "IDH" subtype in Sturm et al. were assigned to G-CIMP subtype in our classification scheme (Figure S5D). The "mesenchymal" tumors were assigned to M1 and M2 (21/25), "RTK II 'classic'" tumors were assigned to M3 and M4 (30/34) and the "RTK I 'PDGFRA'" tumors were assigned to M6. No TCGA samples were clustered in the Sturm et al.'s "G34" or "K27" classes, and we found the corresponding histone mutations to be absent across the TCGA sample set.

mutation if ≥10% transcribed allelic frequency. Deletions are defined by log2 ratios ≤1 or ≤0.5 and focally targeting the gene (see Extended Experimental Procedures). Amplifications are defined by log2 ratio > 2 or > 1 and focal.

(C) Left: for a cohort of 25 GBMs for which whole-genome sequencing allowed genotyping, TERT promoter C228T and C250T mutations occurred in a mutually exclusive fashion. All four TERT promoter wild-type GBM harbored ATRX mutation, and were enriched in G-CIMP group. Right: TERT promoter mutations are associated with elevated expression. Box plots: bar denotes median, central box spans the middle quartiles and whiskers span the full range.

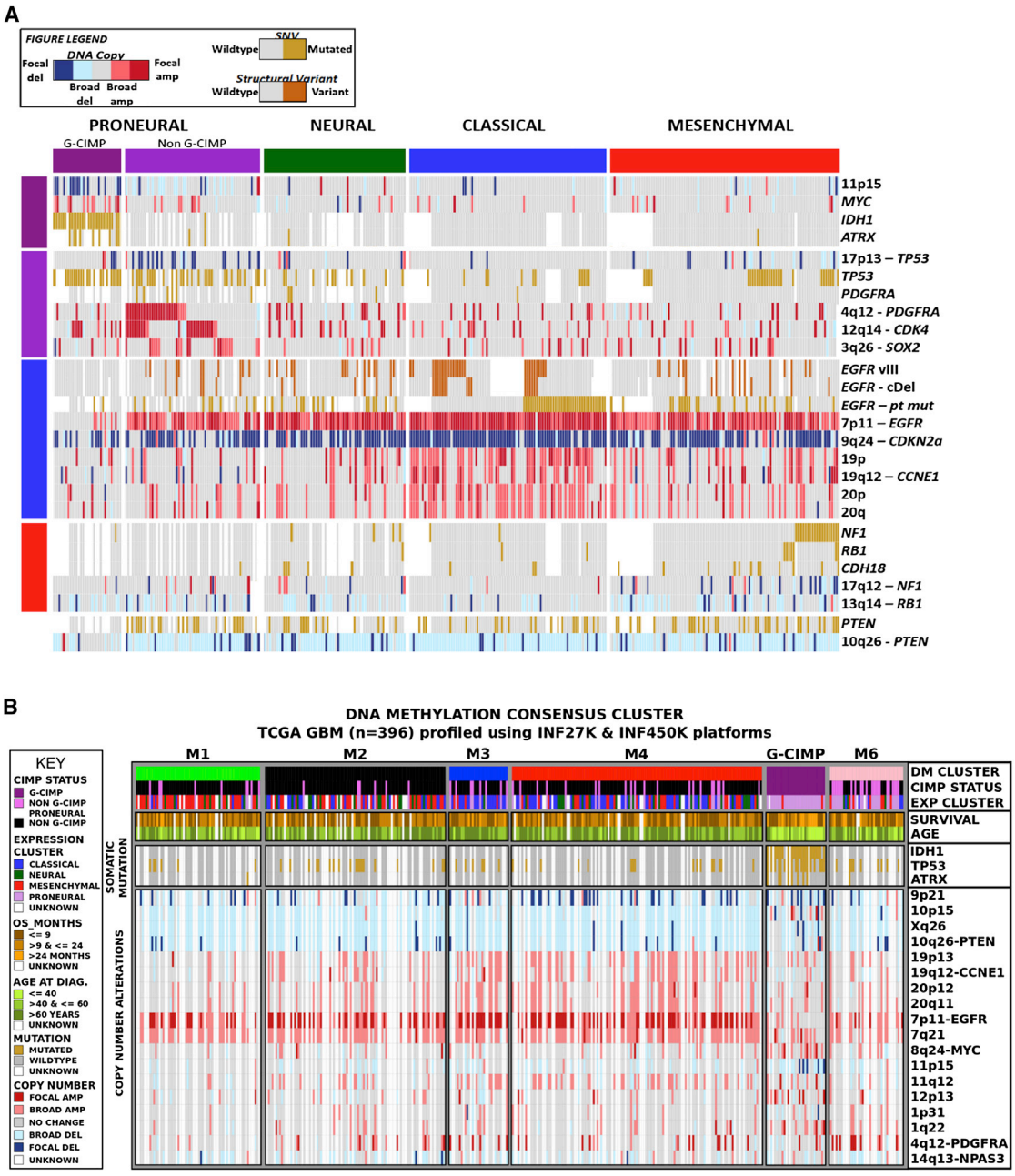


Figure 5. Molecular Subclasses of GBM and their Genomic Molecular Correlates

(A) Genomic alterations and survival associated with five molecular subtypes of GBM. Expression and DNA methylation profiles were used to classify 332 GBMs with available (native DNA and whole-genome amplified DNA) exome sequencing and DNA copy-number levels. The most significant genomic associations were identified through Chi-square tests, with p values corrected for multiple testing using the Benjamini-Hochberg method.

(B) Genomic alterations and sample features associated with six GBM methylation clusters. Epigenomic consensus clustering was performed on 396 GBM samples profiled across two different platforms (Infinium HM27 and Infinium HM450). Six DNA methylation clusters were identified (see related Figure S5), represented as M1 to M6, where M5 is G-CIMP. These DNA methylation signatures are correlated with 27 selected features composed of clinical, somatic, and copy-number alterations; DM cluster, G-CIMP status, four TCGA GBM gene expression subclasses, two clinical features (Age at diagnosis/overall survival in months), somatic mutations (*IDH1*, *TP53*, *ATRX*), and 18 selected copy-number alterations.

Lastly, we explored the relationship of molecular subclasses with clinical parameters such as treatment response or survival. In the current larger TCGA cohort, the survival advantage of proneural subtype GBM (Phillips et al., 2006) was definitively shown to be conferred by G-CIMP status, with non-G-CIMP proneural GBMs and not mesenchymal GBM tending to show less favorable outcomes in the first 12 months following initial diagnosis compared to other subtypes (p value 0.07; Figure S6A). Although

most of the samples clustered in the M6 group were classified as proneural, this methylation subclass was not associated with adverse survival overall (Figure S6B) (Noushmehr et al., 2010).This observation reinforces the notion that target genes affected by the G-CIMP phenotype likely contribute to the improved prognosis for this subset of proneural GBM.

DNA methylation of the *MGMT* gene promoter is a known marker for treatment response (Hegi et al., 2005). We found that the *MGMT* locus was methylated in 48.5% of patients in our cohort (174 of 359 assessed) and that G-CIMP cases showed an increased likelihood of having *MGMT* DNA methylation (79% of G-CIMP versus 46% for non-G-CIMP; Figure S6C). When correlated with outcome, *MGMT* status distinguished responders from nonresponders among samples classified as classical (n = 96; p = 0.01) but not among samples classified as proneural (n = 66; p = 0.57), mesenchymal (n = 104; p = 0.62), and neural (n = 55; p = 0.12) (Figures S6D and S6E). In summary, our data provide evidence for *MGMT* DNA methylation as a predictive biomarker in the GBM Classical subtype of GBM but not other subtypes.

Regulatory Networks of miRNA and mRNA in Gliomagenesis

MicroRNAs (miRs) have been found to promote or suppress oncogenesis through modulation of gene expression via mRNA degradation or inhibition of translation (Bartel, 2004; Krol et al., 2010). Recent studies have proposed additional mechanisms of miR-mRNA regulation, including modulation of competing endogenous RNA (ceRNA), which are mRNA with competitive miR binding sites (Sumazin et al., 2011; Tay et al., 2011). Leveraging the existence of matched mRNA and miR profiling data on a large number of samples, we sought to define the salient interactions between specific pairs of miRs and mRNAs through both of these mechanisms.

We employed a relevance network-based approach to infer miR:mRNA associations in GBMs with matched miR and mRNA profiles (n = 482). Putative regulatory targets of individual miRs were defined as those genes having strong negative correlation with the miR ($\leq$0.3) and prediction support in three commonly used databases (Miranda, Pictar, TargetScan). One hundred and thirty-three miR:mRNA associations defined the final putative miR regulatory network (see Data Portal). The most prevalent associations related to molecular subtypes. For instance, hsa-mir-29a (part of the miR29 family, thought to play a role in the TP53 pathway (Park et al., 2009), was predicted to regulate 23 genes. Seventeen of these 23 genes were expressed at distinctively high levels in the non-G-CIMP+ proneural tumors only and not in the G-CIMP+ tumors. Interestingly, three (*BCL11A*, *PCFG3*, and *SS18L1*) of the 23 genes in this subnetwork are predicted to have competitive binding with PDGFRA regulatory miRs.

ceRNAs are mRNAs coregulated *in trans* by a common miR (Sumazin et al., 2011; Tay et al., 2011). Here, we used a correlation- and NLS-based approach, integrating miRNA and mRNA expression and copy-number profiles to predict ceRNAs for four GBM signature genes: *PDGFRA*, *EGFR*, *NF1*, and *PTEN*. Interestingly, predicted *PDGFRA* ceRNAs significantly overlapped with proneural GBM signature genes (p value $< 1 \times 10^{-15}$), whereas *EGFR* ceRNAs significantly overlapped with classical GBM signature genes (p value = 1.2×10^{-14}) (see Data Portal). Predicted ceRNAs of *NF1* overlapped with proneural signatures ($p < 1 \times 10^{-15}$) and *PTEN*-associated ceRNAs were correlated with the mesenchymal signature. This provocative finding raises the possibility that ceRNA regulation by miR may contribute to the transcriptomic signature that defines the molecular subtypes in GBM, although this hypothesis remains to be tested.

Signaling Pathway Activation in Different Molecular Subtypes of GBM

To assess whether enrichment of genomic alterations in molecular subtypes translates into downstream pathway activation, we performed targeted proteomic profiling by RPPA. Two hundred and fourteen sample lysates were probed with 171 antibodies targeting phospho- and/or total-protein levels among signaling pathways as previously described (TCGA, 2012c). After normalization, coclusters of correlated signaling molecules within specific signaling pathways were observed (see Extended Experimental Procedures; Data Portal) and were utilized as readout of pathway activity status for correlative analyses.

Unsupervised clustering of RPPA data failed to produce a consistent partitioning of the sample cohort into clearly-defined subtypes. However, 127 out of the 171 antibodies were found to correlate significantly with transcriptomal subtype (Kruskal-Wallis, $p < 0.05$; Extended Experimental Procedures). As anticipated, *EGFR* amplification/mutation was associated with significant elevations in total EGFR expression ($p = 3.74 \times 10^{-15}$) and phosphorylation ($p = 1.44 \times 10^{-12}$, Figure S7A), both prominent in classical subtype tumors (Figure S7B). Classical GBMs also showed relative downregulation of proapoptotic proteins (including cleaved caspase 7, cleaved caspase 9, Bid, and Bak) as well as MAP kinase signaling, including its downstream target p90RSK. Notch1 and Notch3 expression were moderately increased in classical tumors, consistent with previous reports linking EGFR and Notch activation in GBM (Brennan et al., 2009).

Mesenchymal subtype tumors exhibited elevated levels of endothelial markers, such as CD31 and VEGFR-2, consistent with previous findings (Phillips et al., 2006), as well as markers of inflammation (e.g., fibronectin and its downstream target COX-2). Mesenchymal tumors showed moderately increased activation of the MAPK pathway, as evidenced by higher levels of phospho-Raf, phospho-MEK, and phospho-ERK (Figure 6). These tumors also exhibited decreased levels of the mTOR regulatory protein, tuberin (*TSC2* gene product), which is inhibited by ERK phosphorylation.

In contrast to the mesenchymal subtype, proneural GBMs showed relatively elevated expression and activation of the PI3K pathway including the Akt-regulated mTorc1 activation site (Figure 6). Proneural tumors showed greater inhibition of the 4EBP1 translation repressor, whereas mesenchymal tumors display elevated S6 kinase activation (indicative of mTOR effector pathway activation). Therefore, both subtypes achieve mTOR pathway activation although the specific patterns of steady-state protein activation differ.

G-CIMP+ tumors shared characteristics with their proneural superfamily, but also showed decreased expression of several proteins, including Cox-2, IGFBP2, and Annexin 1. Among the 171 antibodies tested in the TCGA data set, these three proteins

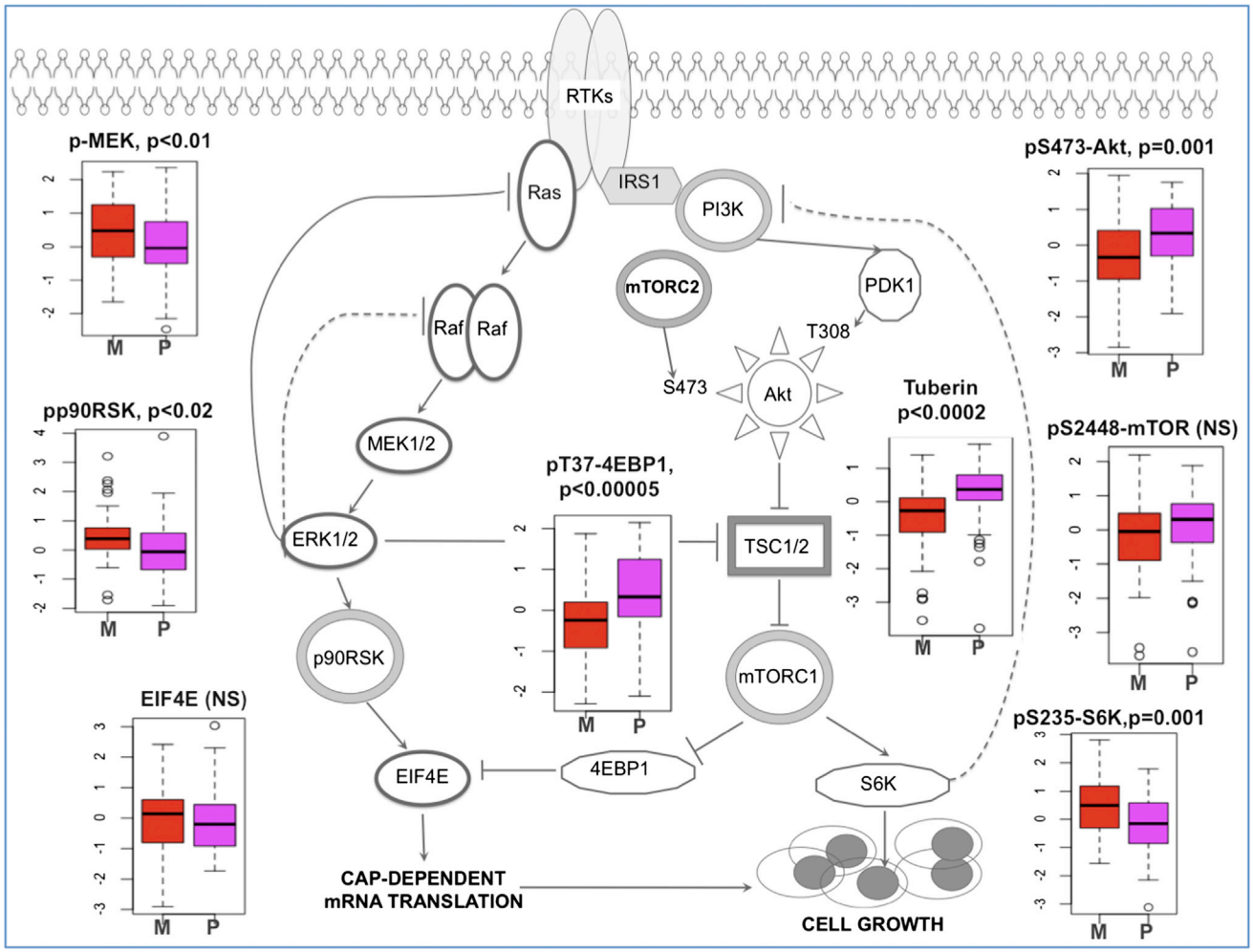


Figure 6. Canonical PI3K and MAPK Pathway Activation Determined by Reverse Phase Protein Arrays and Compared between GBM Subclasses

Proneural (P, purple, n = 55) and mesenchymal (M, red, n = 45). Activation/expression levels are plotted for principal signaling nodes of the MAPK (phospho-MEK and phospho-p90RSK), PI3 kinase (pS473-Akt) and mTOR (TSC1/2, phospho-mTOR, p235/236 S6, phospho-4EBP1 and EIF4E) pathways (p values, two-tailed t test). Mesenchymal tumors showed increased activation of the MAPK pathway (evidenced by higher levels of phospho-MEK and downstream phospho-p90RSK) and decreased levels of phospho-ERK inhibitory target TSC2. In contrast, proneural tumors showed relatively elevated expression and activation of members of the PI(3) kinase pathway including Akt PDK1 target site threonine 308 (p = 0.01, data not shown) and Akt mTORC2 target site (serine 473). Phospho-ERK levels were not significantly different between these two subtypes. Box plots: bar denotes median, central box spans the middle quartiles and whiskers denote extremes up to 1.5 time the middle interquartile range.

were the most negatively prognostic (Cox proportional hazard test, $p < 0.0004–0.0013$). IGFBP2 and Cox-2 have been independently reported as poor prognostic markers in diffuse gliomas (Holmes et al., 2012; Shono et al., 2001), and low IGFBP2 expression has been associated with global DNA hypermethylation in glioma (Zheng et al., 2011). Members of the annexin family have been associated with glioma growth and migration, and annexin-1 is known to be underexpressed in secondary but not primary GBM (Schittenhelm et al., 2009). Together, the correlations of these proteins with G-CIMP status suggest that their prognostic significance is not independent. Analysis of DNA methylation for *IGFBP2*, *COX2*, and *ANXA1* found no evidence of hypermethylation in G-CIMP tumors.

Interestingly, samples with RTK amplification had lower levels of canonical RTK-target pathway activities as measured by phospho-AKT, phospho-S6 kinase, and phospho-MAPK cocluster levels (Figure S7C). Whereas *PTEN* loss and deletion were each associated with incremental increases in AKT pathway activity, PI3K mutant samples had lower AKT activity than samples lacking *PI3K* mutations, concordant with findings in breast cancer (TCGA, 2012c). Samples harboring *NF1* mutation/deletion showed elevated MAP kinase activity (p-ERK and p-MEK, p value < 0.001), and trended toward decreased PKC pathway activity. These examples of nonlinear relationship between protein signaling and underlying genetic mutations speak to complex and likely dynamic signaling in cancers.

DISCUSSION

In this study, we provided a comprehensive catalog of somatic alterations associated with glioblastoma, constructed through whole-genome, exome, and RNA sequencing as well as copy-number, transcriptomic, epigenomic, and targeted proteomic profiling. With the availability of detailed clinical information including treatment and survival outcome for nearly the entire cohort, this rich data set offers new opportunity to discover genomics-based biomarkers, validate disease-related mechanisms and generate novel hypotheses.

In addition to alterations in signature oncogenes of GBM, such as *EGFR* and *PI3K*, we found that over 40% of tumors harbor at least one nonsynonymous mutation among the chromatin-modifier genes. A role for chromatin organization in GBM pathology, which has been described for cancer types such as ovarian carcinoma (Wiegand et al., 2010) and renal carcinoma (Varela et al., 2011), is suggested. We also detected mutations in genes for which targeted therapies have been developed, such as *BRAF* (Chapman et al., 2011a), and *FGFR1/FGFR2/FGFR3* (Singh et al., 2012), demonstrating the potential clinical impact of this TCGA data set.

Structural rearrangements that contributed to the overall complexity of the genome and transcriptome were detected in the majority of GBM. A high frequency of structural variants on the q arm of chromosome 12, involving the *MDM2* and *CDK4* genes, was observed and associated with the presence of double minute, extrachromosomal DNA fragments, which may be functionally relevant (Zheng et al., 2013). The identification of complex *EGFR* fusion and deletion variants in nearly half of GBM confirm relevance of this category of somatic alterations to the disease. While the development of a therapeutic strategy targeting mutated *EGFR* could have a major impact on survival and continues to be a topic of great interest (Vivanco et al., 2012), strategies will need to address the possibility that different *EGFR* alterations might exist concurrently in a tumor and yield differential biological activities and/or responses to any given targeted inhibitor.

Another level of biological complexity is revealed by targeted proteomic profile, which showed that the impact of specific genomic alterations on downstream pathway signaling is not linear and not always predictably concordant with genotype. This observation has provocative clinical implication as it directly challenges the notion that therapeutic inhibition of downstream signaling components along a pathway would yield similar efficacy of targeting the mutated gene itself. Additionally, this observation highlights the limitation of TCGA data, namely its inherent static nature given a single time point analysis, and its inability to map specific genetic or protein changes to the individual cells or cell populations through whole-tumor tissue analysis.

In summary, this report reaffirms the power and value of TCGA's comprehensive multidimensional and clinically annotated GBM reference data set in enabling hypothesis generation based on unanticipated observations and relationships emerged from unbiased data-driven analyses. We believe that this public resource will serve to facilitate discovery of new insights that can advance our molecular understanding of this disease.

EXPERIMENTAL PROCEDURES

Patient and Sample Characteristics

Specimens were obtained from patients, with appropriate consent from institutional review boards. Details of sample preparation are described in the Extended Experimental Procedures.

Data Generation

In total, 599 patients were assayed on at least one molecular profiling platform, which platforms included: (1) exome sequencing, (2) DNA copy-number and single-nucleotide polymorphism arrays, (3) whole-genome sequencing, (4) gene expression arrays, (5) RNA sequencing, (6) DNA methylation arrays, (7) reverse phase protein arrays, and (8) miRNA arrays. Details of data generation are described in the Extended Experimental Procedures.

Whole-Genome and Exome Sequencing Data Analysis

Massively Parallel Sequencing Exome capture was performed by using Agilent SureSelect Human All Exon 50 Mb according the manufacturer's instructions. All exome and whole-genome sequencing was performed on the Illumina GA2000 and HiSeq platforms. Basic alignment and sequence quality control were done by using the Picard and Firehose pipelines at the Broad Institute. Mapped genomes were processed by the Broad Firehose pipeline to perform additional quality control, variant calling, and mutational significance analysis.

RNA Sequencing Data Analysis

Libraries were generated from total RNA and constructed using the manufacturers protocols. Sequencing was done on the Illumina HiSeq platform. Read mapping and downstream data analysis (expression profiles, fusion transcripts, and structural transcript variants) were performed using the PRADA pipeline.

Array Data Preprocessing and Analysis

To ensure across-platform comparability, features from all array platforms were compared to a reference genome as previously described (TCGA, 2008). Both single platform analyses and integrated cross-platform analyses were performed, as described in detail in the Extended Experimental Procedures.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Extended Experimental Procedures, six figures, and seven tables and can be found with this article online at http://dx.doi.org/10.1016/j.cell.2013.09.034.

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Update

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miR-9a Minimizes the Phenotypic Impact of Genomic Diversity by Buffering a Transcription Factor

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In the above article, we did not indicate a Sequence Read Archive (SRA) accession number for the DNA sequence data obtained to draw our conclusions. These data are available under the accession number SRP039006 at http://www.ncbi.nlm.nih.gov/sra/.

The Somatic Genomic Landscape of Glioblastoma

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In the above article, we stated that our SNV mutations count included 5,379 silent, 3,901 missense, 831 nonsense, 360 splice-site, and 760 mutations resulting in a frame shift. One of these figures represents an error; the missense mutation count should read 13,901 and not 3,901. The underlying data as well as the rest of the text and figures reflect the correct mutation count (13,901).